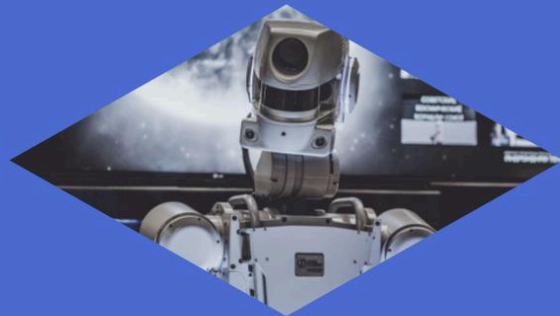


Интегрированная теория информации: способ измерения сознания в искусственном интеллекте?

Автор: Саша Кадариу / Обновлено 24 августа 2022 г.

Integrated Information Theory: A Way to Measure Consciousness in AI?

We explain the 'Hard' Problem, the essential properties of experience, the Phi Metric and the application of IIT to Evaluate Consciousness in AI, and more.



Понятие **сознательного ИИ** широко исследовалось во всей научной фантастике, от сериала Айзека Азимова "Я робот" до современных фильмов, таких как "Из машины", "Бегущий за лезвием 2049", и трилогии "Матрица". В некоторых случаях в мейнстриме возможность сознательного искусственного интеллекта даже связывали с разработкой [агентов широкого профиля](#) и [чат-ботов](#). В то время как СМИ рисуют картину мира, в котором синтетическое/искусственное сознание коренным образом меняет представление о том, что значит быть человеком, они не особо задумываются о тех самых элементах, которые порождают сознательный опыт.

Люди ценят свое существование, потому что оно имеет для них значение. Каждый опыт, который мы приобретаем в течение жизни, влияет на нашу индивидуальность. Опыт позволяет нам соглашаться или не соглашаться с другими, формировать систему убеждений и личные идеологии, а также решать, как мы хотим прожить свою жизнь. Он влияет на то, как мы мыслим, учимся и развиваемся с течением времени. Наша способность **воздействовать на мир и подвергаться его влиянию** — это то, что придает жизни структуру, и именно в рамках этой структуры мы можем находить смысл.

Тем не менее большинство из нас воспринимает тот факт, что мы **обладаем сознанием**, как нечто само собой разумеющееся именно благодаря нашей способности придавать смысл вещам через опыт. Наше понимание собственного сознания самоочевидно и неоспоримо. Философ-экзистенциалист XVII века **Рене Декарт** выразил эту идею в своей знаменитой цитате: *«Я мыслю, следовательно, я существую»*.

Однако в случае с искусственным интеллектом мы сталкиваемся с проблемой: как мы можем узнать, **на что это похоже** — думать так, как думает ИИ? И какие логические и физические механизмы в структуре мышления ИИ [привели бы к возникновению сознательного опыта](#)? Даже если бы мы смогли **феноменологически свести** сознательный опыт ИИ к его сущностным свойствам или квалиа, смогли бы мы соотнести эти свойства с сознанием и понять их?

Интегративная теория информации (ИТИ), разработанная нейробиологом Джулио Тонони, призвана ответить на эти вопросы универсальным и уникальным с точки зрения математической оценки способом. Но прежде чем мы перейдем к ИТИ, давайте вкратце обсудим «сложную» проблему.

«Сложная» проблема

Наука — это описательная и исследовательская область, основная цель которой — объяснить свойства и законы, управляющие нашей физической реальностью, опираясь на эмпирические данные и наблюдения. Цель наблюдения в рамках научного метода — *уточнить, опровергнуть или адаптировать существующие теоретические концепции*. Например, **ньютоновская механика** основана на идее о том, что гравитация — это сила, прямо пропорциональная массе объекта. Однако только после того, как Эйнштейн разработал свою **теорию относительности**, мы поняли, что это не совсем так: гравитация — это искривление пространства-времени. Эта концепция была подтверждена многочисленными астрономическими наблюдениями, такими как принцип [гравитационного линзирования](#).

Gravity is **observable and quantifiable**; it is one of the fundamental properties that govern our existence in the natural world – it can be explained and justified by reference to physical phenomena alone. Consciousness, however, is a **metaphysical property** of human existence that manifests itself through **subjective experience**. While there must necessarily be physical systems that give rise to it, attempting to root our conception of consciousness in these systems presents us with an intractable problem: how can we know [what it is like](#) to experience the world as something else? For instance, if I show you the color red, the area known as the ‘color center’ in your ventral occipital lobe will activate, revealing the physical system that is responsible for your capacity to perceive color. However, my subjective experience of the color red will **intrinsically differ** from yours – I might find that it inspires feelings of anger while you find it to be relaxing.

The Hard Problem of consciousness can thus be summarized in the following way: *while we are capable of understanding the physical systems that underlie the basic elements of our cognition, we cannot, by reference to these systems and the capacities they give rise to adequately explain why it is that something is having a conscious experience, to begin with*. This

would necessitate an additional **first-person qualitative** and functional understanding of what it is like to experience the world from the perspective of the entity in question.

IIT thus aims to develop a theory of consciousness that is empirically and quantitatively grounded but retains the ability for qualitative assessment.

Essential Properties of Experience or 'Axioms'

IIT considers an unusual approach to the characterization of consciousness by adopting a **top-down theoretical framework**. Instead of trying to identify various **neural correlates** and functionally explain how it is that they give rise to conscious experiences, IIT assumes that the essential properties of consciousness are **self-evident**. In other words, the theory seeks to first characterize the nature of consciousness and then define its origins.

IIT posits five '**axioms**' or *essential properties* of conscious experience:

1. Axiom of existence
2. Axiom of composition
3. Axiom of information
4. Axiom of integration
5. Axiom of exclusion

The axiom of existence stems from the initial **cartesian assumption**, which is encapsulated in the Descartes quote mentioned earlier, that the **existence and experience of consciousness are undeniable**. When an entity is conscious, it perceives its existence from its perspective, its experiences are **immediate and actual** – the reality of these experiences cannot be denied by external observers.

The axiom of composition posits that all conscious experience is in some way **structured**, and this structure is what allows us to distinguish between various elements in our physical reality. For instance, my capacity to perceive physical pain or pleasure will in turn structure my

physical experience and behavior, namely the extent to which I seek out objects or experiences that relate to these phenomena.

The axiom of information goes one step further, purporting the idea that experience is not universal, but rather **specific and targeted**. My experience of kicking a ball is **non-transferable**; it is distinct from my experience driving a car, watching a movie, or cooking a meal.

The axiom of integration, which is at the heart of IIT, states that the characteristics of conscious experience are not mutually exclusive, but rather **interdependent** – *consciousness is holistic and unified by nature*. For instance, if I see a red fire hydrant, I cannot separate my experience of the color red from my identification of the object as a fire hydrant. The experience is unified; what I see before me is precisely a red fire hydrant.

Finally, the axiom of exclusion states that conscious experience is **definite** both in terms of content and the speed at which it occurs. My experience of an empty dark room, while it is distinguishable from my experience of my bedroom, is not actually any more or less potent; *consciousness is specified*, and as an effect of this specificity it is also exclusive, meaning that each conscious experience is equally potent. Moreover, the rate at which conscious experience occurs is consistent – the **temporal flow** of conscious experience remains constant across experiential domains.

From Axioms to Postulates

Postulates within IIT are what justify and describe axioms or the essential features of conscious experience in terms of their **physical properties**. Importantly, physical systems must be composed of **elements in a state** whereby each element must possess at least **two internal states**. When an input is provided it affects the internal states of an element in a specific fashion, such that the output produced is a function of the change in internal states. **Neurons** and **logic gates** represent some common examples in this context. Understanding what is meant by

elements in a state is vital to the notion of cause and effect, which functionally underpins all the physical properties of axioms.

The axiom of existence claims that the experience of consciousness is intrinsic; the substrates of the system that gives rise to conscious experience must then be composed of elements in a state that are themselves **intrinsic to the system**. For such a system to exist it must be **actionable**, effectively possessing **cause-effect power**; *a thing cannot exist if it cannot act or be acted upon*. The axiom of existence does not merely necessitate that consciousness retains the function of causal power, but also that it can exercise cause and effect upon itself, precisely because it is self-referential and subjective.

The axiom of composition claims that conscious experience is structured; this implies that any subsets of elements in the physical substrate must be able to come together to form a **meaningful composition** or representation of conscious phenomena. Thus, the compositions formed between various physical elements of a conscious system must also possess cause-effect power; if this is the case, then elemental compositions allow for the development of **higher order mechanisms**, which contribute to and uphold the complex structure of conscious experience.

The axiom of information claims the specificity of conscious experiences. If an experience is distinct, then the systematic structure that gives rise to it must itself possess mechanistic properties that constitute what IIT refers to as a **cause-effect repertoire**; this repertoire is what allows for the **differentiation of specific experiences**, effectively constituting a system's **cause-effect structure**. At any point in time, this structure is in a given state – the cause-effect repertoire defines and constitutes the cause-effect power of any mechanism in a given state, while the cause-effect structure defines and constitutes the cause-effect power of every mechanism and its corresponding system of elements.

The axiom of integration claims that consciousness is holistic and unified. This implies that the mechanical and functional elements composing the system must be **interdependent**; every mechanistic element must retain the power to exercise cause over other elements and be affected by individual elements as well as the system as a whole. In other words, if a system can be **structurally reduced** such that individual elements are removed incrementally without manipulating or changing the overall cause-effect structure, then the system in question does not satisfy this postulate.

The axiom of exclusion claims that conscious experience is definite and temporally consistent. This implies that the subsets or supersets of mechanistic elements in a system have **differential cause-effect structures**, but of these subsets or supersets, only one is a maximally irreducible conceptual structure, or as IIT refers to it, **MCIS**. In other words, a conglomeration of intertwined cause-effect structures would make no difference to the system; if a mechanism is in a given state such that it specifies a specific cause-effect repertoire concerning one system, it should additionally specify another cause-effect repertoire for another system. This would be **redundant**.

These postulates, taken in conjunction with their respective axioms, posit that consciousness is a system whose functional properties can be defined by the aforementioned postulates while its phenomenology can be characterized by reference to its axioms.

The Phi Metric and the Application of IIT to Evaluate Consciousness in AI

The ultimate goal of IIT is to evaluate and mathematically quantify the degree of conscious experience any system has. This is achieved by calculating the maximum amount of **integrated information** in a system, which is quantified using the **phi metric**. For instance, a computer's hardware is composed of logic circuits, which define the connections between respective logic gates or elements in a state. Suppose we were

to systematically partition the various mechanistic elements in a system, and find that no connections are lost. In that case, we can conclude the system cannot exercise **cause-effect power upon itself**, thereby having no phi value.

Similarly, a system where partitioning leads to the severing of some connections but not others would have a comparatively **low phi value**. In simpler terms, if we find that the removal or activation of certain elements does not affect others, we can assume the system has a **low degree of connectivity**, and subsequently, information integration. The more holistic a system is, the higher its phi value; if the elements composing a system are **non-redundant**, interdependent, and universally connected, then we can expect to record a high phi value. Importantly, phi does not evaluate the quality of conscious experience, but rather, its **quantity**, whereas the axioms are what provide phenomenological and qualitative insight.

While IIT is, quite frankly, a beautifully succinct and versatile theory of consciousness that can be applied in both synthetic and biological systems, it is nonetheless limited in cases where **dimensionality and complexity** increase exponentially, under the principle of [Bell's number](#). In a relatively simplistic system with a limited number of elements, the number of possible partitions can still be obtained pragmatically. However, as systems become more complex, effectively increasing the number of elements that constitute respective mechanisms, the number of connections between elements will increase exponentially. This makes the calculation of the phi metric, while still possible in principle, extremely difficult. Consider the human brain, which has over 86 billion neurons; how could one possibly begin partitioning the various systems in such a complex structure? The same issue applies to [large neural network](#) and NLP models, some of which currently have over 10 million nodes and up to 176 billion parameters.

Fortunately, the pragmatic limits of calculating phi could theoretically be resolved by implementing **computational models** whose primary function is the partitioning of various elements within a system. With the aid of

computer processing power, it may be possible to calculate a phi value within highly complex systems. If a high phi value is obtained, and the relevant phenomenological axioms specified by IIT are also identified within a system, then it is likely that the system possesses some degree of consciousness. However, as with most theories of consciousness, caution should be taken, especially when attempting to classify synthetic or artificial systems as the functionality and methodology of their cognition differs from our own.

Nonetheless, IIT predicts consciousness in systems with a sufficiently high level of information integration – the question we should be considering, in this case, is whether phi should have a **threshold value**; if certain systems possess mechanistic elements that correspond with the relevant axioms and cause-effect structure but lack a unified composition, we need to be able to evaluate whether a **minimum level of connectivity** is sufficient for conscious experience. It would be wise, in our attempts to qualitatively and quantitatively assess consciousness in AI systems, to generate a comparative model that we can use to assess the degree of consciousness in AI. I would intuitively assume that such a model would be based on the human brain since it is currently the only system we know for sure is capable of conscious experience.



Sasha Cadariu

Sasha is an AI Strategist, Researcher, and Consultant with a deep interest in applied ethics, focusing on the potential impact of AI-driven exponential technologies on humanity.

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